2018 Ph H2 Q5

Section: Our Dynamic Universe

Topic: Relativity

A spacecraft is travelling past Earth with a constant speed of $0.90c$, where $c = 3.00 \times 10^8 \text{ m s}^{-1}$.
An astronaut on the spacecraft measures the time taken for a signal to travel between two detectors placed a certain distance apart.

(a) State the term used to describe the time interval measured by the astronaut.

✅ This is called the **proper time**.

Proper time is the time **measured in the frame of reference where the two events occur at the same location** — in this case, the astronaut's frame.

(b) The proper time measured by the astronaut is 8.0×10^{-6} s.

Calculate the time interval measured by an observer on Earth.

Use the **time dilation** formula:

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where:

- $t_0 = 8.0 \times 10^{-6}$ s (proper time)
- $v = 0.90c$

$$t = \frac{8.0 \times 10^{-6}}{\sqrt{1 - (0.90)^2}} = \frac{8.0 \times 10^{-6}}{\sqrt{1 - 0.81}} = \frac{8.0 \times 10^{-6}}{\sqrt{0.19}}$$

$$t = \frac{8.0 \times 10^{-6}}{0.4359} = 1.83 \times 10^{-5} \text{ s}$$

🧠 Revision Tips

- **Proper time** is measured by the observer at rest relative to the two events (i.e. where both events happen at the same place).
- Use time dilation for **relativistic speeds**:
$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$
- Time **dilates**: moving clocks run **slower** relative to stationary observers.
- Always express speeds as a **fraction of** c when using relativity equations.
